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Title: Enhancing Audit Efficiency through Artificial Intelligence: A Case Study in Financial Reporting

IEEE Citation

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Summary:

This paper explores how artificial intelligence (AI) enhances audit efficiency in financial reporting, using a detailed case study of a multinational corporation's audit process. The authors argue that AI technologies, particularly machine learning and natural language processing, streamline repetitive audit tasks, improve data analysis accuracy, and reduce audit completion times. By focusing on a single firm's adoption of AI tools, the study provides practical insights into how AI can transform financial audits, addressing challenges like high data volumes and tight regulatory deadlines.

The methodology centers on a case study of a global manufacturing company (2019-2023), supplemented by quantitative metrics on audit performance and qualitative feedback from audit team members. The AI system implemented included automated data extraction from financial statements and predictive models for identifying high-risk transactions. Results show a 32% reduction in audit time and a 15% decrease in error rates, with specific improvements in detecting inconsistencies in revenue recognition. The case study also highlights challenges, such as initial resistance from auditors unfamiliar with AI tools and the need for robust data governance to ensure compliance with standards like SOX.

The findings suggest that AI not only boosts efficiency but also allows auditors to focus on complex, judgment-based tasks, enhancing overall audit quality. However, the authors note barriers like integration costs and the need for ongoing training to align AI with human expertise. They advocate for phased AI adoption and clear documentation of AI processes to build trust among stakeholders. The paper concludes by emphasizing AI's potential to redefine audit workflows, calling for broader studies to validate these findings across diverse industries.